Reflection

Trent Gwathney | IS601 | Docker Compose and PostgreSQL

Adding PostgreSQL and pgAdmin to the FastAPI Calculator helped me understand how a web application depends on several services working together. Docker Compose provides a repeatable configuration for those services. One detail that became clearer was the difference between using localhost in my browser and using db as the database hostname inside another container.

The users and calculations tables made one-to-many relationships easier to understand. One user can have several calculations, while each calculation belongs to one user. The foreign key enforces that relationship, and ON DELETE CASCADE removes associated calculations when their user is deleted. I also paid closer attention to WHERE clauses because an update or deletion without one can affect every record. Changing the stored result to 6 demonstrated an update, but also showed why application-level validation matters.

One setup challenge was Docker Desktop failing to start because of a stale socket. That reminded me to distinguish an application problem from an environment problem. The calculator's 130 automated tests passed, but that alone did not prove the containers or SQL operations worked. Separate Compose and database checks provided that evidence.

I feel more comfortable organizing a Compose project, understanding persistent volumes, and writing basic SQL queries. Keeping database records in a named volume also clarified why stopping a container should not erase its data. I still want more practice troubleshooting container networking, writing more complex joins, and using transactions to make database changes safely.